

Database Management Systems – Assignment

Q1. What is SQL?

Ans: SQL, short for Structured Query Language, is a standardized programming language specifically designed for storing, managing, and retrieving data held in a relational database. It was developed in the early 1970s at IBM (originally called SEQUEL) and later became the ANSI/ISO standard language for interacting with relational database systems. Almost every relational DBMS - MySQL, Oracle, SQL Server, PostgreSQL understands SQL, with only minor syntax variations between them.

SQL lets a user or an application describe 'what' result they want, rather than 'how' to get it - the database engine itself figures out the most efficient way to fetch or modify the data. Because of this, SQL is called a declarative language.

Key characteristics of SQL:

- **Easy to learn:** syntax is close to plain English (SELECT, WHERE, ORDER BY, etc.)
- **Non-procedural:** you specify the desired outcome, not the step-by-step logic
- **Standardized:** works (with minor dialect differences) across almost all relational databases
- **Powerful:** a single query can filter, sort, group, and join data from multiple tables at once.

SQL is broadly grouped into four categories of commands:

Category	Full Form	Purpose	Example Commands
DDL	Data Definition Language	Defines database structure	CREATE, ALTER, DROP, TRUNCATE
DML	Data Manipulation Language	Works with the data itself	SELECT, INSERT, UPDATE, DELETE
DCL	Data Control Language	Manages access/permissions	GRANT, REVOKE
TCL	Transaction Control Language	Manages transactions	COMMIT, ROLLBACK, SAVEPOINT

Q2. Difference between Data and Information?

Ans: Data is the raw material, unorganized facts with no context. Information is what you get once that data has been processed, arranged, and given meaning so it can support understanding or a decision.

Basis	Data	Information
Meaning	Raw, unprocessed facts	Processed and meaningful facts
Form	Numbers, words, symbols	Organized, structured content
Dependency	Independent unit	Depends on data
Usefulness	Not directly useful for decisions	Useful for decision-making
Example	21, "Raj", 90	"Raj, aged 21, scored 90 marks"

Q3. What is a Database and a DBMS?

Ans: A **database** is an organized, electronically stored collection of related data — for instance, a school's records of its students, subjects, and marks.

A **DBMS (Database Management System)** is the software layer that sits between the user and the database. Instead of a user reading or writing raw files directly, every request passes through the DBMS, which handles storage, retrieval, security, backup, and consistency.



Fig. 1 — How a user interacts with data through a DBMS

Q4. What is a Relational Database Management System (RDBMS)?

Ans: An RDBMS is a DBMS built around the relational model, where data is stored in tables (also called relations). Each table has rows (records) and columns (attributes), and tables can be linked to one another using keys — a primary key uniquely identifies each row in a table, and a foreign key in one table refers to a primary key in another, creating a relationship between them.

Example — a Students table and an Enrollments table linked by StudentID:

StudentID (PK)	Name	Age
101	Raj	21
102	Meena	20

Students table (PK = Primary Key)

Enroll_ID	Student_ID (FK)	Course
1	101	DBMS
2	102	Data Structures

Enrollments table — StudentID here is a Foreign Key referring to the Students table

RDBMS software also enforces normalization (reducing data redundancy) and the ACID properties — Atomicity, Consistency, Isolation, Durability — which guarantee that transactions complete reliably even in cases of failure or concurrent access.

Q5. At Least 5 Database Management Systems / Software.

DBMS	Type	Developed By
MySQL	Relational (Open Source)	Oracle Corporation
Oracle Database	Relational	Oracle Corporation
Microsoft SQL Server	Relational	Microsoft
PostgreSQL	Relational (Open Source)	PostgreSQL Global Dev. Group
MongoDB	NoSQL (Document-based)	MongoDB Inc.
SQLite	Relational (Embedded)	D. Richard Hipp